Knowledge Representation

by Adia Khalid

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Lecture notes by Liliana Sanfilippo

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Part I

Foundation of Knowledge Representation

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Introduction to KR: Definition, Scope, Motivation (Use Cases), and History

1.1 Introduction

What is Knowledge Representation?

Knowledge Representation

Definition 1.1 KR is a central problem in Artificial Intelligence that focuses on developing sufficiently precise notations for representing knowledge

The core problem with KR is that the real world with its complexity, ambiguity and dynamism can never be fully represented in a computer-based system. Nevertheless, intelligent systems must be able to function effectively using partial, incomplete and sometimes contradictory information.

Why is Knowledge Representation needed?

The problem of data meaninglessness

To understand the need for KR, let's consider a practical scenario:

Example 1.1 A student database.
A hypothetical database contains the following entries:

Name	**Major**	**CGPA**	**Research**	...
Alice Smith	CS	3.8	Genomics	...
Bob John	Biology	3.8	Oncology	...
Carol Li	Zoology	3.7	Animal colonisation	...

Without knowledge representation, this data is disjointed. The entries exist in isolation across thousands of files. A database query system can only perform exact keyword matches:

- Query 1: "Show me all AI researchers" → No results (not explicitly mentioned in the data)
- Query 2: "Which students could work on this patent?" → No answer possible
- Query 3: "What are the emerging trends?" → No analysis possible

Data is meaningless without context and relationships. It does not allow for intelligent reasoning or deeper analysis.

What knowledge representation makes possible

With knowledge representation, we can structure the data and transfer isolated records into interconnected, queryable knowledge using:

- ontologys: Explicitly specify which concepts exist (student, researcher, field of research) and how they are categorised.
- Knowledge graphs: Link entities together: Alice → researcher → AI

This allows for semantic searches, automatic relationship discovery and reasoning across all connection.

Example 1.2
- Semantic search
  - "Find students researching machine learning?"
  - "Show me all AI researchers": The system can use transitive relationships to find relevant students.
  - "Which students are a good fit for this patent?": The system can match skills and areas of interest.
- Automatic relationship discovery
  - "What are emerging research trends?"
  - Recognising patterns across all data
  - Identifying emerging trends through interconnected information

Data	Information	Knowledge
Raw, unprocessed facts	Structured and contextualised data	Information combined with experience and reasoning rules
e.g. CGPA: 3.8	e.g. Alice has a CGPA of 3.8 in Computer Science	e.g. Alice could work on this AI patent because she has a strong foundation in computer science and is interested in genomics

Table 1.1: Data vs. Information vs. Knowledge

KR enables machines to make informed decisions with fewer rules.

KR Perspective

Representation Schemes

Challenges in Knowledge Representation

Traditional ML systems follow a black-box approach which means the decision-making process is not transparent and the reasoning behind predictions is not accessible.

- Generalisation: Requires continuous retraining with new data
- Explainability: Difficult or impossible to explain decisions
- Data flexibility: Highly dependent on large amounts of training data

Knowledge-Based AI utilises structured, explicit knowledge.

explicit knowledge

Definition 1.2 Knowledge that is formally encoded in symbols, rules or graphs and can be processed by machines

- Generalization: explicit knowledge transfers to new problems without retraining from scratch[2]
- Explainability: encoded knowledge enables transparent reasoning and human-understandable decisions[1]
- Data flexibility: Modifiable

Representation Schemes

representation scheme

Definition 1.3 A representation scheme is a formal notation used to define a knowledge base (Mylopoulos 1980). A knowledge base contains facts and rules and serves as a model of the real world. A representation scheme is therefore the 'language' in which we express knowledge, much like programming languages define the syntax and semantics in which we write instructions.

Example 1.3 **Pac-Man ghost AI** has a knowledge-based approach: We code ghost behaviour as explicit rules.

1. IF player_nearby THEN chase
2. IF frightened THEN flee
3. ELSE patrol

Decisions are transparent, explainable, and modifiable without recomputing. (Ogland 2016)

Challenges in Knowledge Representation

Knowledge representation involves designing formal structures that allow computers to store, interpret, and reason about information and every design decision has advantages and caveats.

Choosing the representation is one of the design challenges. Semantic networks are intuitive and visually understandable but have limited inference. First-order logic on the other hand is expressive, but has computationally costly reasoning.

Desired properties can conflict. Systems should ideally be expressive enough to capture rich real-world knowledge while remaining efficient enough to reason in practical time. Tradeoffs are between expressiveness and tractability, and Human readability and machine efficiency.

Application-driven constraints arise from, different domains requiring different forms of representation. E.g. In natural language processing (NLP), systems must handle ambiguity and context. In medical diagnosis, representations must manage uncertainty, causal relationships, and strict correctness requirements. In question answering, systems need fast retrieval combined with reasoning over large amounts of structured and unstructured data

The core idea of Knowledge Representation is not just a data structure. It plays multiple roles simultaneously and those roles can conflict with each other:

- Surrogate for the world: Acts as a substitute for real world objects and relationships
- Ontological commitments: Defines what exists and how things are categorized
- Theory of reasoning: Provides foundation for logical inference and deduction
- Medium for efficient computation: Enables performant computational processing
- Medium of Human expression: Allows humans to express knowledge in understandable ways

1.2 Taxonomy of Representation Schemes

[3]

1.3 Distinguishing Features of KR Systems

Defis: Frame, Inference,

kinds of reasoning inference types (23+) David, Shrobe and Szolovist

Wie man mit Incompleteness umgeht

1.4 Applications of Knowledge Representation

Enterprise Knowledge Management

1.5 Future Directions

Multi-hop reasoning Unified framework

knowledge Aggregation

def attention weight

1.6 Course structure

foundation of KR

Knowledge graphs

KR in mL

grading:

50% of the exercise points (full portfolio) (Abgabe in 3er Gruppen(?)) present one exercise in tut present one paper in seminar 40–45 min (3er gruppen) final report 5-.10 pages (50% marks on final report)

Es wird ein Moodle geben

1.7 Formalities

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Logical Foundations: First-Order Logic (FOL)

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Logic Programming and Automated Reasoning

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Structured Knowledge Representation

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ontology todo. 8

representation scheme A representation scheme is a formal notation used to define a knowledge base (Mylopoulos 1980). A knowledge base contains facts and rules and serves as a model of the real world. A representation scheme is therefore the 'language' in which we express knowledge, much like programming languages define the syntax and semantics in which we write instructions.. 9

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Bibliography

[1] Jiahang Cao et al. "Knowledge Graph Embedding: A Survey from the Perspective of Representation Spaces". In: *ACM Comput. Surv.* 56.6 (Mar. 2024). ISSN: 0360-0300. DOI: 10.1145/3643806. URL: https://doi.org/10.1145/3643806.

[2] Aidan Hogan et al. "Knowledge Graphs". In: *ACM Comput. Surv.* 54.4 (July 2021). ISSN: 0360-0300. DOI: 10.1145/3447772. URL: https://doi.org/10.1145/3447772.

[3] *Knowledge Representation 01 - Knowledge Basic Introduction, Motivation, History*. Vol. 1. Lecture Notes Knowledge Representation.